

Decision Tree

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import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier, plot_tree

# Convert categorical data to numerical
X = pd.get_dummies(data.iloc[:10, :-1]) # Use first 10 records
y = data.iloc[:10, -1].map({'Yes':1, 'No':0})

# Build ID3 Decision Tree
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

# Plot the tree
plt.figure(figsize=(18,10))
plot_tree(model,
          feature_names=X.columns,
          class_names=['No', 'Yes'],
          filled=True,
          rounded=True)

plt.title("Decision Tree using ID3 Algorithm")
plt.show()
```

